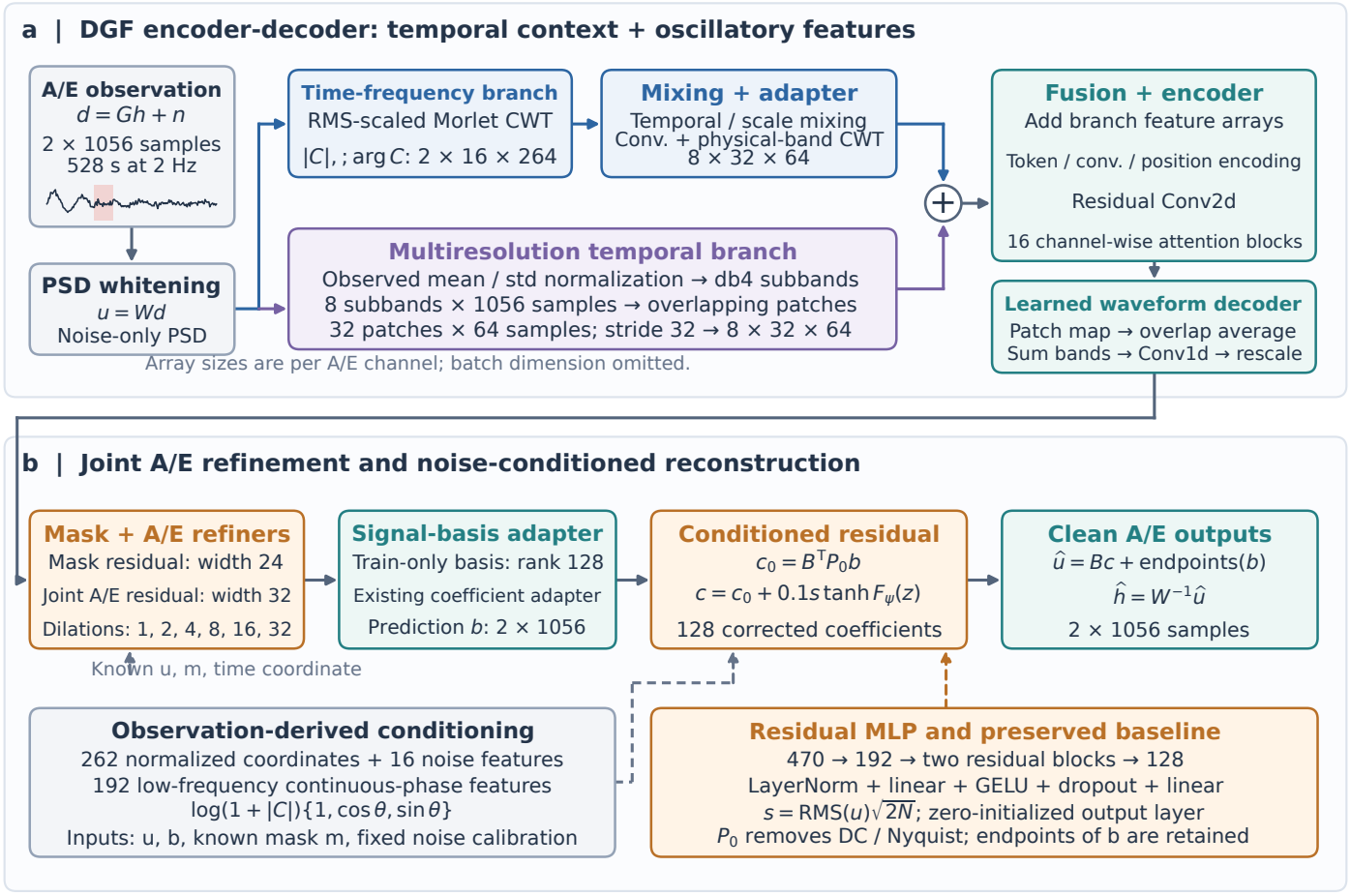
a | DGF encoder-decoder: temporal context + oscillatory features
A/E observation
$d = Gh + n$
2 × 1056 samples
528 s at 2 Hz
PSD whitening
$u = Wd$
Noise-only PSD
Time-frequency branch
RMS-scaled Morlet CWT
$|C|, ; \arg C$: 2 × 16 × 264
Mixing + adapter
Temporal / scale mixing
Conv. + physical-band CWT
8 × 32 × 64
Multiresolution temporal branch
Observed mean / std normalization → db4 subbands
8 subbands × 1056 samples → overlapping patches
32 patches × 64 samples; stride 32 → 8 × 32 × 64
Fusion + encoder
Add branch feature arrays
Token / conv. / position encoding
Residual Conv2d
16 channel-wise attention blocks
Learned waveform decoder
Patch map → overlap average
Sum bands → Conv1d → rescale
Array sizes are per A/E channel; batch dimension omitted.
b | Joint A/E refinement and noise-conditioned reconstruction
Mask + A/E refiners
Mask residual: width 24
Joint A/E residual: width 32
Dilations: 1, 2, 4, 8, 16, 32
Known u, m, time coordinate
Signal-basis adapter
Train-only basis: rank 128
Existing coefficient adapter
Prediction b: 2 × 1056
Conditioned residual
$c_0 = B^{\mathsf{T}} P_0 b$
$c = c_0 + 0.1 s \tanh F_\psi(z)$
128 corrected coefficients
Clean A/E outputs
$\hat{u} = Bc + \text{endpoints}(b)$
$\hat{h} = W^{-1}\hat{u}$
2 × 1056 samples
Observation-derived conditioning
262 normalized coordinates + 16 noise features
192 low-frequency continuous-phase features
$\log(1 + |C|)\{1, \cos\theta, \sin\theta\}$
Inputs: u, b, known mask m, fixed noise calibration
Residual MLP and preserved baseline
470 → 192 → two residual blocks → 128
LayerNorm + linear + GELU + dropout + linear
$s = \text{RMS}(u)\sqrt{2N}$; zero-initialized output layer
P_0 removes DC / Nyquist; endpoints of b are retained